🧱 Phase 1: Data Engineering & Quality Control (Summary)

In this phase, I focused on optimizing the database structure and implementing rigorous data quality checks. This process transformed raw data into a reliable "source of truth" by eliminating errors that could have distorted business results.

1. Import Optimization & Schema Safety

- **Redundant Data Removal:** During the data ingestion phase, I eliminated unnecessary columns to reduce the database footprint and improve query performance.
- **Data Type Precision:** I modified the customer_zip_code_prefix column to VARCHAR(20). This is a crucial step to preserve leading zeros and prevent incorrect numerical formatting (such as thousand separators) in postal codes.
- **Storage Efficiency:** I manually adjusted VARCHAR lengths to ensure data integrity while optimizing storage performance.

2. Text and Financial Standardization (Data Integrity)

- **Geographical Normalization:** In the customers and sellers tables, I normalized city names to uppercase using the UPPER function. This ensured accurate geographical data aggregation by removing inconsistencies caused by different letter cases.
- **Currency Accuracy:** Within the created SQL Views, I enforced a two-decimal format (DECIMAL(10,2)). This prevents rounding errors typical of floating-point formats and ensures that revenue totals are perfectly accurate.

3. Advanced Duplicate Resolution (Window Functions)

- **Problem Identification:** In the order_reviews table, I identified duplicates where rows shared the same review_id but had different timestamps.
- **The Solution:** I implemented a robust cleaning strategy using the ROW_NUMBER() window function (partitioned by review_id and ordered by date descending). This ensured that only the most recent and unique record for each review was preserved, maintaining the integrity of customer satisfaction metrics.

4. Documentation of SQL Views (The "Gold" Layer)

I created a view layer to separate raw data from the analytical layer. These views automatically prepare data for visualization:

- **v_payments_pretty:** Cleans payment method names (replacing underscores with spaces) and standardizes the currency format.
- **v_category_translation:** Maps technical category names to readable English translations.
- **v_products_pretty:** Prepares a product list with sanitized names, enabling efficient grouping and catalog analysis.

5. Comprehensive NULL Audit & Data Pruning

- **Audit Results:** I performed a system-wide audit to identify missing values. In the products table, I detected 610 records with missing category names and two records with zero physical dimensions.
- **Data Cleaning:** These records were systematically excluded from the analysis to prevent "Unknown" categories from skewing business insights.

6. Future Process Optimization

- **Strategy:** While I performed all transformations in SQL to maintain full control over the database logic, I plan to leverage **Power Query (M language)** for future large-scale projects.
- **Benefits:** Using visual profiling tools like "Column Quality" will allow for faster error detection and more automated data preparation during the final visualization phase.